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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025 (Old Course)

Subject: Mathematics - II (Differential Equation)

Code: 101202

Time: 03 Hours

Full Marks: 70

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) For the differential equation, $(x-1)d^2y/dx^2+\cot(\pi x)dy/dx+ (\operatorname{cosec}^2\pi x)y = 0$ which of the following statement is true?

(i) 0 is regular and 1 is irregular
(ii) 0 is irregular and 1 is regular
(iii) Both 0 and 1 are regular
(iv) Both 0 and 1 are irregular

(b) Let $P_n(x)$ be the Legendre polynomial of degree $n$ge0. Then $\int_{-1}^{1}P_n(x)\,dx = 2$ if n is

(i) 0
(ii) 1
(iii) 2
(iv) None of these

(c) $\int_0^{\pi/2}\sqrt{(\pi x)}J_{1/2}(2x)\,dx = k$ then k is

(i) 0
(ii) 1
(iii) 2
(iv) $\sqrt{(\pi)}$

(d) The solution of $p+q=z$ is

(i) $f(xy, y\log z) = 0$
(ii) $f(x+y, y+\log z) = 0$
(iii) $f(x-y, y-\log z) = 0$
(iv) $f(x/y, y/\log z) = 0$

(e) The complementary function of $[\partial^2z] / [\partial x^2]- 4[\partial^2z] / [\partial x\partial y]+ 4[\partial^2z] / [\partial y^2]= x+y$ is

(i) $f(y+2x) + xg(y+2x)$
(ii) $f(y-2x) + xg(y-2x)$
(iii) $x\,f(y+2x) + y\,g(y+2x)$
(iv) $x\,f(y+2x) - y\,g(y+2x)$

(f) The transformation w =sin$z$ is conformal

(i) at all points
(ii) at all points except $z = n\pi+\pi/2$, n=0,±1,±2,dots
(iii) at all points except $z = n\pi+\pi/2$, n=0, 1, 2,dots
(iv) at all points except $z = n\pi+\pi/2$, n=1, 2, 3,dots

(g) The value of $\int_C[3z^2+ 7z + 1] / [z-4]dz$, where C is $9x^2+ 4y^2= 36$ is

(i) $2\pi i$
(ii) 0
(iii) $3\pi i$
(iv) $4\pi i$

(h) Newton's iterative formula to find the value of $\sqrt{(N)}$ is

(i) $x_{n+1}=1/2(x_n- N/x_n)$
(ii) $x_{n+1}=1/2(x_n- Nx_n)$
(iii) $x_{n+1}=1/2(x_n+ N/x_n)$
(iv) $x_{n+1}=1/2(x_n+ Nx_n)$

(i) While applying Simpson's 3/8 rule the number of sub intervals should be

(i) Odd
(ii) 8
(iii) Even
(iv) Multiple of 3

(j) In the geometrical meaning of Euler's algorithm, the curve is approximated as a

(i) Straight line
(ii) Circle
(iii) Parabola
(iv) Ellipse

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Q.2 Solve both questions : **[14]**

(a) Solve $y = 2px + y^2p^3$.

(b) Solve $(px-y)(py+x) = a^2p$.

Q.3 Solve both questions : **[14]**

(a) Find the solution of the differential equation $x^2y'' - 4xy' + 6y = x^4\cos x$, $y(\pi)=0$, $y'(\pi)=1$.

(b) State and prove the orthogonal property of Legendre polynomial.

Q.4 Solve both questions : **[14]**

(a) Find the solution of the following partial differential equation: $(x^2-yz)p + (y^2-zx)q = z^2-xy$

(b) Find the solution of the following partial differential equation: $2z + p^2 + qy + 2y^2 = 0$

Q.5 Solve both questions : **[14]**

(a) Find the solution of the partial differential equation $(D - 5D' - 6)z = 6e^{6x}\log(y+5x)$.

(b) Show that the function $f(z) = \sqrt{(|xy|)}$ is not analytic at the origin even though C.R. equations are satisfied thereof.

Q.6 Solve both questions : **[14]**

(a) Find the Taylor series expansion of the function $f(z) = \sin z$. Also find the values of $f^{(2n)}(0)$ and $f^{(2n-1)}(0)$.

(b) Write the statement of Cauchy Integral formula. Hence, evaluate the integral $\int_C [z-1] / [(z+1)^2(z-2)]dz$ where C is the circle $|z-i|=2$.

Q.7 Solve both questions : **[14]**

(a) Find the value of the integral $\int_0^\infty [\cos ax - \cos bx] / [x^2]dx$ where $a, b > 0$.

(b) Discuss the rate of convergence of Bisection method.

Q.8 Solve both questions : **[14]**

(a) A third degree polynomial passes through the points (0,1), (1, 1), (2, 1) and (3, -2). Find the polynomial.

(b) Evaluate $\int_4^{5/2} \log x \, dx$ by Simpson's 1/3 rule and Simpson's 3/8 rule, by dividing the range into 6 parts.

Q.9 Solve this question : **[14]**

Using Euler's modified method, find numerical solution of the differential equation $dy/dx = x + |\sqrt{(y)}|$ with $y(0)=1$ for $0 \le x \le 0.6$, in steps of 0.2.

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